

BATCH No:26

**VOICE-ENABLED OBJECT DETECTION FOR THE
VISUALLY IMPAIRED USING CNN**

*Major project report submitted
in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology
in
Information Technology**

By

B. LEELA KRISHNA MOHAN (21UTIT0009) (VTU19580)

*Under the guidance of
Ms. J. DEEPA, B.E., M.E.,
ASSISTANT PROFESSOR*



**DEPARTMENT OF INFORMATION TECHNOLOGY
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE AND TECHNOLOGY**

(Deemed to be University Estd u/s 3 of UGC Act, 1956)

**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

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CERTIFICATE

It is certified that the work contained in the project report titled "VOICE-ENABLED OBJECT DETECTION FOR THE VISUALLY IMPAIRED USING CNN" by B. LEELA KRISHNA MOHAN (21UTIT0009) has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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May, 2025

DECLARATION

We declare that this written submission represents my ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

(Signature)

(B.LEELA KRISHNA MOHAN)

Date: / /

APPROVAL SHEET

This project report entitled VOICE-ENABLED OBJECT DETECTION FOR THE VISUALLY IM-PAIRED USING CNN by B.LEELA KRISHNA MOHAN (21UTIT0009) is approved for the degree of B.Tech in Information Technology.

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Place:

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ABSTRACT

This project presents an innovative approach to object detection using Convolutional Neural Network (CNN) algorithm integrated with voice output. Traditional object detection systems primarily rely on visual outputs, which may not be accessible to visually impaired individuals or may require continuous visual attention from users. Our proposed system utilizes CNN, a powerful deep learning technique for image recognition and classification, to accurately detect objects within images. The system is augmented with voice output capabilities, enabling it to audibly announce the detected objects, thus making the information accessible to visually impaired users or in scenarios where visual attention is not feasible. Moreover, the system allows users to input data files containing images for object detection, enhancing its versatility and usability. By accepting data files as input, users can seamlessly integrate the object detection system into their existing workflows and processes. The combination of CNN algorithm, voice output, and data file input enhances the accessibility, usability, and adaptability of object detection systems, making them more inclusive and versatile across various applications and user scenarios. Furthermore, we explore the integration of voice output functionality, describing how the detected objects are converted into audible announcements. The system's voice output capabilities are designed to be clear, concise, and easily understandable, facilitating seamless interaction for users. Additionally, we elaborate on the implementation of the data file input feature, which allows users to provide input images for object detection through files rather than real-time capture. This feature enhances flexibility and convenience, particularly in scenarios where batch processing or integration with existing data pipelines is required. We evaluate the performance of the proposed system through extensive experiments, assessing its accuracy, speed, and robustness across various datasets and scenarios.

Keywords: Visually impaired, Blindness, Assistive system, Computer vision, Image recognition, Voice output, Convolutional Neural Network, Audio.

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LIST OF ACRONYMS AND ABBREVIATIONS

Acronym	Abbreviation
AOD	Assistive Object Detection
CNN	Convolution Neural Network
ODAI	Object Detection for the Visually Impaired
ODVI	Object Detection for Visual Impairment
ODI	Object Detection for Impaired

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Chapter 1

INTRODUCTION

1.1 Introduction

India being an agriculture country, about 70 percent of the population depends on it as their main source of income and food. Agriculture plays an important part of the Indian economy as it contributes about 17 percent of the total GDP. Farmers have a wide range in selecting their crops and finding a suitable pesticide for it but in spite of all their efforts it can all be vain if they can't identify the disease plaguing their crops. Thus, disease on crops can significantly reduce the quality and quantity of agricultural products along with economical damage to the farmers. To successfully cultivate crops without incurring much loss we need to properly identify the disease and remedy it, this requires a lot of work and processing time as detecting each and every plant can be tedious and time consuming. To lessen the burden of the farmers along with their losses we propose the use of a system which can detect infected plants so that we can curb the spread of infection and diseases at an earlier step thus reducing losses and crop failure.

To tackle the problem of early and accurate detection of plant diseases, our system utilizes advanced image processing techniques combined with deep learning algorithms, specifically Convolutional Neural Networks (CNNs). CNNs have shown remarkable success in image classification tasks and are well-suited for identifying disease patterns from plant leaf images. By training the model on a dataset of various plant diseases, our system learns to distinguish between healthy and infected leaves with high accuracy. Once the disease is detected, the system can notify the user with the name of the disease and suggest possible remedies. This not only saves time but also enables even non-experts to recognize and act upon early signs of infection without needing in-person assistance from agricultural experts.

The implementation of this system can be done using a mobile or web application where farmers can upload pictures of their crops. The system then processes the image, detects any disease symptoms, and delivers a result almost instantly. This

makes it convenient and highly accessible, even in remote areas with limited access to agricultural support. The system can also be updated over time with new disease data to improve its accuracy and scope. By maintaining a centralized database of common plant diseases and their symptoms, the system remains relevant and effective as agricultural conditions evolve due to climate change and other external factors.

1.2 Background

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1.3 Objective

The scope of a project for object detection aimed at assisting impaired individuals encompasses various aspects. Firstly, it involves researching and selecting appropriate technologies such as computer vision and machine learning algorithms tailored to detect and recognize objects. Hardware considerations like cameras, sensors, and processing units need to be evaluated for effectiveness and compatibility. The project must include the development of a user-friendly interface accessible to people with diverse impairments, ensuring usability and inclusivity. Integration with existing assistive technologies or devices may also be necessary for seamless interaction. 2 Data collection and annotation are crucial phases, involving the creation of comprehensive datasets containing diverse objects in various environments to train and validate the detection models effectively. Testing and validation procedures should cover a wide range of scenarios to assess the system's performance, accuracy, and robustness in real-world conditions, accounting for factors like lighting conditions, object occlusion, and user mobility. Deployment strategies need to be devised considering scalability, sustainability, and maintenance requirements. Furthermore, the project should address ethical and privacy considerations, ensuring user consent, data security, and compliance with relevant regulations. Documentation, training materials, and support mechanisms must be developed to facilitate the adoption and utilization

of the system by impaired individuals and their caregivers. Continuous improvement through user feedback and technological advancements should be integrated into the project's lifecycle to ensure its long-term efficacy and relevance. User interface design will focus on intuitive interaction methods, such as voice commands or tactile inputs, to relay object detection results effectively. Usability testing with target users will validate system performance and refine user experience. The project aims to enhance accessibility and independence by providing reliable object recognition capabilities, ultimately empowering visually impaired individuals to navigate their surroundings with greater confidence and safety.

1.4 Problem Statement

Object detection systems have become pivotal in numerous applications, ranging from autonomous vehicles to surveillance and accessibility tools. However, traditional systems predominantly rely on visual interfaces, making them less effective or completely inaccessible to individuals with visual impairments. Additionally, these systems often require constant visual attention, which may not be feasible in certain real-world scenarios, such as while driving or operating machinery. The absence of alternative output modalities limits the inclusivity and usability of such systems, especially for users who depend on auditory information for interaction.

Furthermore, most object detection implementations are designed for real-time use through cameras, lacking the flexibility to process static data files. This limitation hinders integration with existing workflows where batch processing or offline analysis of images is necessary. There is a growing need for a more adaptable solution that not only delivers high accuracy in object detection but also offers accessible output through audio and allows data file input for increased convenience. Addressing these gaps can significantly improve the usability and reach of object detection systems across diverse user needs and operational contexts.

Chapter 2

LITERATURE REVIEW

Yang, R., et.al.[13] This paper provided an overview of computer vision techniques tailored for assisting visually impaired individuals, including object detection methods and their applications. The survey on computer vision for visually impaired individuals aims to gather insights into their needs and preferences regarding assistive technology. Questions cover usage patterns, desired features, and barriers faced when accessing visual information.

Lee, S. et al.[7] The paper proposed a real-time object detection system based on deep learning for providing auditory feedback to the visually impaired. Real-time object detection for blind individuals using deep learning involves employing convolutional neural networks (CNNs) to process live video feeds from cameras or sensors. The system identifies objects in the user's surroundings and provides auditory or tactile feedback to convey their presence and location.

Wang, Y. et al.[12] This work presented a wearable device that integrates object detection and recognition capabilities to assist visually impaired users in navigating their surroundings. A wearable assistive device for object detection and recognition for visually impaired individuals typically consists of a small camera or sensor array mounted on glasses or a headband. This device utilizes computer vision algorithms, often based on deep learning, to analyze the visual input in real-time.

Chen, X. et al.[3] The authors explored methods to improve object recognition accuracy using smartphone cameras, catering specifically to the needs of visually impaired users. Enhancing object recognition for the visually impaired involves refining algorithms for accuracy, speed, and adaptability to diverse environments. Integration of multi-sensory feedback, customization options, and continuous improvement mechanisms ensures more intuitive and effective assistance, empowering users to navigate their surroundings with greater confidence and independence.

Zhang, L. et al.[15] This paper presented a system that leverages convolutional neural networks (CNNs) for real-time object recognition and auditory feedback for visually impaired users.

Liu, M. et al.[8] The study investigated efficient object detection techniques suit-

able for wearable devices, aiming to optimize performance and resource utilization. Additionally, feedback on existing solutions and suggestions for improvement are solicited. The survey seeks to inform the development of more effective and user-friendly computer vision systems for the visually impaired community.

Gupta, et al.[4] This research proposed an assistive object detection system utilizing Internet of Things (IoT) technology and deep learning for assisting visually impaired individuals. This technology enhances the independence and mobility of visually impaired individuals by enabling them to navigate and interact with their environment more effectively.

Li, et al.[6] The paper explored the feasibility of real-time object detection on edge devices, offering low-latency feedback suitable for visually impaired users. It identifies objects in the user's surroundings and provides auditory or tactile feedback to convey information about their presence, location, and characteristics. The device enhances the user's independence and mobility by allowing them to navigate their environment interact with objects more effectively.

Martinez, P. et al.[9] This work presented an accessible object detection system implemented on embedded platforms to provide practical support for blind and visually impaired individuals. A real-time object recognition system for impaired individuals employs computer vision algorithms to detect objects in their environment instantly.

Yang, Q. et al.[14] The study evaluated different object detection models for wearable devices, analyzing their effectiveness in assisting visually impaired users. It enhances independence and navigation for visually impaired individuals, enabling them to perceive and interact with their surroundings more effectively.

Ahmed et al. [1] proposed an Edge-AI enabled solution in the form of smart glasses that provide real-time plant disease detection and guidance. This wearable system is capable of processing data locally, thus offering fast and context-aware agricultural insights without relying heavily on cloud infrastructure.

Bashir et al.[2] developed a CNN-based real-time object detection system integrated with speech synthesis, aiming to assist visually impaired users. Their work demonstrates how combining computer vision and audio feedback can enhance spatial awareness and interaction for blind individuals.

Kumar and Sharma [5] took a voice-driven approach by implementing a voice-assisted plant disease recognition system that leverages deep learning models in conjunction with IoT devices. Their model allows users, especially those with limited literacy or technical skills, to interact with the system using natural speech, thus

making agricultural diagnostics more accessible.

Mehta et al. [10] introduced a smart assistive system tailored for crop disease identification. Their system not only detects disease but also provides actionable guidance to farmers, highlighting the potential of intelligent tools in enhancing agricultural productivity and decision-making.

Nair et al.[11] focused on the development of an audio-visual feedback system specifically designed for visually impaired individuals. By utilizing deep learning techniques, their system provides real-time feedback through both sound and visual cues, improving the user's ability to perceive and navigate their environment. Collectively, these studies underscore the growing impact of AI-based assistive technologies in domains such as agriculture and accessibility, where real-time feedback, user-friendly interaction, and reliable detection are critical.

2.1 Existing System

The existing systems for object detection designed to assist impaired people employ various technologies such as computer vision and machine learning to identify and classify objects in the environment. These systems typically utilize cameras integrated into smartphones or wearable devices to capture images, which are then processed in real-time to detect objects. The detected objects are conveyed to the user through auditory or haptic feedback, providing crucial information about the surroundings to aid navigation and awareness.

However, existing systems may still face several disadvantages. One common challenge is the accuracy and reliability of object detection, especially in complex or cluttered environments. False positives or missed detections can occur, leading to incorrect or incomplete information being relayed to the user. Additionally, the computational resources required for real-time object detection can strain the capabilities of mobile devices, impacting the system's responsiveness and efficiency. Another limitation is the dependency on camera-based input, which may be affected by lighting conditions or occlusions, potentially hindering the system's performance in certain situations. Lastly, the usability and accessibility of these systems can vary based on user preferences and technological proficiency, emphasizing the importance of user-centered design and continuous improvement in this domain.

Despite these challenges, ongoing research and development efforts continue to advance object detection systems for impaired individuals, aiming to enhance reli-

ability, accuracy, and user experience through innovative technologies and methodologies.

2.2 Related Work

Over the years, object detection has evolved significantly, with Convolutional Neural Networks (CNNs) becoming the backbone of most state-of-the-art detection models. Prominent models such as YOLO (You Only Look Once), SSD (Single Shot MultiBox Detector), and Faster R-CNN have demonstrated high accuracy and real-time performance in detecting and classifying objects in images and video streams. These models have been widely applied in areas like autonomous driving, surveillance, and mobile applications. In parallel, assistive technologies for the visually impaired, such as Microsoft's Seeing AI and Google's Lookout, have integrated object recognition with voice output to improve accessibility. These systems generally rely on real-time camera input and mobile device integration to provide feedback through audio cues.

2.3 Research Gap

While existing systems have made significant progress in object detection and accessibility, several gaps remain. First, many assistive tools are limited by their dependence on real-time camera input, which restricts their use in scenarios where batch processing or offline image analysis is needed. Second, few systems provide an option to input image data through files, limiting integration into automated workflows or existing databases. Third, although some applications offer voice feedback, the customization, clarity, and flexibility of the voice output often remain underdeveloped, reducing usability for users with specific auditory needs. Thus, there is a need for a system that combines robust CNN-based object detection with flexible image input (including file-based input) and high-quality, customizable voice output to enhance accessibility, usability, and integration in various real-world scenarios.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

The existing systems for object detection designed to assist impaired people employ various technologies such as computer vision and machine learning to identify and classify objects in the environment. These systems typically utilize cameras integrated into smartphones or wearable devices to capture images, which are then processed in real-time to detect objects. The detected objects are conveyed to the user through auditory or haptic feedback, providing crucial information about the surroundings to aid navigation and awareness.

However, existing systems may still face several disadvantages. One common challenge is the accuracy and reliability of object detection, especially in complex or cluttered environments. False positives or missed detections can occur, leading to incorrect or incomplete information being relayed to the user. Additionally, the computational resources required for real-time object detection can strain the capabilities of mobile devices, impacting the system's responsiveness and efficiency. Another limitation is the dependency on camera-based input, which may be affected by lighting conditions or occlusions, potentially hindering the system's performance in certain situations. Lastly, the usability and accessibility of these systems can vary based on user preferences and technological proficiency, emphasizing the importance of user-centered design and continuous improvement in this domain.

Despite these challenges, ongoing research and development efforts continue to advance object detection systems for impaired individuals, aiming to enhance reliability, accuracy, and user experience through innovative technologies and methodologies.

3.2 Proposed System

The proposed system for object detection aimed at assisting impaired individuals leverages cutting-edge computer vision techniques and user-centered design principles to overcome existing challenges and provide enhanced functionality. This system integrates advanced object detection algorithms, such as deep learning models like Faster CNN, to accurately identify and classify objects in real-time. By harnessing the power of neural networks, the proposed system can achieve high levels of accuracy and robustness, minimizing false positives and ensuring reliable feedback to the user. One of the key advantages of the proposed system is its emphasis on usability and accessibility. The user interface is designed with visually impaired individuals in mind, featuring intuitive interaction methods such as voice commands or tactile inputs to relay object detection results effectively. This focus on user-centered design enhances the system's usability and adoption, empowering impaired individuals to navigate their surroundings with increased confidence and independence. Additionally, the proposed system can be deployed on various platforms, including smartphones or wearable devices, ensuring versatility and adaptability to different user preferences and needs.

Another advantage of the proposed system is its potential for continuous improvement and adaptation through machine learning. By incorporating adaptive learning techniques, the system can evolve over time based on user feedback and environmental data, further enhancing its performance and reliability. This adaptability enables the system to address diverse scenarios and challenges faced by impaired individuals in real-world environments, ultimately contributing to improved accessibility and quality of life for users.

3.3 Feasibility Study

The proposed system is technically feasible due to the maturity and accessibility of the technologies involved. Convolutional Neural Networks (CNNs) have been widely adopted for object detection tasks and are supported by numerous open-source libraries such as TensorFlow, PyTorch, and OpenCV. These tools provide pre-trained models and easy-to-use frameworks that reduce the complexity of implementation while maintaining high performance. Similarly, integrating voice output can be achieved using text-to-speech (TTS) engines such as Google Text-to-Speech,

Amazon Polly, or open-source options like pyttsx3, which support clear and customizable audio output. The system can be designed to accept images through file input, which simplifies integration with existing data workflows and supports batch processing, making the solution versatile for different operational environments.

From an economic and operational standpoint, the project is also feasible. The required hardware—standard computers with moderate GPU capabilities—makes it cost-effective for small-scale or individual use, while also scalable for larger deployments. The software stack is mostly open-source, minimizing licensing costs. Furthermore, the system’s ability to function offline (after initial setup) enhances its usability in low-connectivity environments, which is particularly beneficial for visually impaired users in remote areas. Given these factors, the proposed solution is both technically and economically viable and holds strong potential for real-world deployment and further development.

3.3.1 Economic Feasibility

The economic feasibility of implementing object detection systems for impaired individuals involves evaluating both initial investment costs and long-term benefits. Initial costs include hardware (such as smartphones or wearable devices with cameras), software development, and training data for machine learning models. However, these costs can be mitigated by leveraging existing technologies and open-source resources.

The long-term benefits of object detection systems for the visually impaired justify the investment. These systems promote independence and safety, potentially reducing the need for constant human assistance. They can also enhance productivity and employment opportunities for visually impaired individuals by improving their ability to navigate and interact with their environment. Overall, the economic feasibility of object detection systems for the impaired is promising, with potential cost savings in the long run and significant improvements in quality of life and inclusion.

3.3.2 Technical Feasibility

The technical feasibility of implementing object detection for impaired individuals is well-established due to advancements in computer vision and machine learning. Object detection algorithms, such as SSD (Single Shot MultiBox Detector), can accurately identify and classify objects in real-time, even on resource-constrained de-

vices like smartphones or wearable gadgets. These systems leverage deep learning techniques and are capable of running efficiently on various platforms, providing auditory or haptic feedback to users about their surroundings. Furthermore, the availability of open-source libraries and pre-trained models simplifies the development and deployment process, making object detection solutions accessible for developers aiming to create assistive technologies for visually impaired individuals.

3.3.3 Social Feasibility

The social feasibility of object detection systems for impaired individuals is highly favorable, as these technologies contribute to greater inclusivity and independence for people with visual impairments. By enhancing accessibility and mobility, object detection systems empower individuals to navigate their environments more confidently and participate more fully in daily activities. These systems also promote social integration by reducing barriers to communication and interaction, fostering a more inclusive society. Additionally, the development and adoption of such technologies demonstrate societal progress towards accommodating diverse needs and promoting equal opportunities for individuals with disabilities. Overall, object detection for impaired people aligns with broader social goals of inclusivity and equity, enhancing the quality of life and social participation of visually impaired individuals.

3.4 System Specification

3.4.1 Hardware Specification

- CPU: Intel Core i5 or AMD Ryzen 5
- GPU: NVIDIA GTX 1060 or equivalent (6GB VRAM)
- RAM: 16GB DDR4
- Storage: 500GB SSD
- OS: Windows 10 or Linux (Ubuntu, CentOS)
- Deep Learning Framework: TensorFlow, PyTorch (with GPU support)

3.4.2 Software Specification

- Operating System: Windows 10 (64-bit) or Linux (for compatibility and driver support)
- Deep Learning Framework: TensorFlow or PyTorch (GPU-accelerated deep learning development)
- Programming Language: Python (3.6 or later for framework compatibility and libraries)
- Development Environment: Jupyter Notebook, Visual Studio Code, or PyCharm (code editing, debugging, visualization)
- Additional Libraries: OpenCV (image processing), NiBabel (neuroimaging data), Scikit-learn (machine learning utilities), Matplotlib/Seaborn (visualization)
- Version Control: Git (code management and collaboration)
- Cloud Computing Platforms (optional): Google Colab, AWS, Azure (pre-configured environments and hardware)
- Data Handling Tools: BraTS dataset tools or custom scripts (for loading and preprocessing)
- Model Evaluation Metrics: Dice similarity coefficient, sensitivity, specificity (brain tumor segmentation performance assessment)

3.4.3 Tools and Technologies Used

- **Python** – Used as the primary programming language due to its extensive support for machine learning and image processing libraries.
- **TensorFlow / Keras** – Deep learning frameworks used to build and train the Convolutional Neural Network (CNN) for plant disease detection. Keras provides a simple API while TensorFlow handles backend computation.
- **OpenCV** – Utilized for various image processing tasks such as resizing, denoising, and enhancing images before feeding them into the CNN model.
- **NumPy and Pandas** – Employed for efficient numerical operations, data preprocessing, and dataset management.

- **Matplotlib / Seaborn** – Used to visualize training and validation results, such as accuracy/loss graphs and confusion matrices.
- **Jupyter Notebook / Google Colab** – Provided an interactive environment for developing and testing the deep learning model, with GPU support for faster training.
- **Labelling / Annotation Tools** – Used for labeling and annotating images when creating or refining custom datasets.
- **Scikit-learn** – Utilized for data splitting, model evaluation, and generating classification reports including precision, recall, and F1-score.
- **Text-to-Speech (TTS) Libraries** – Libraries such as `pyttsx3` or `gTTS` were used to convert detected results into audible speech for enhanced accessibility.
- **Flask / Streamlit (Optional)** – Lightweight frameworks used to build a user interface allowing users to upload images for real-time plant disease detection.

3.4.4 Standards and Policies

Anaconda Prompt

Anaconda prompt is a type of command line interface which explicitly deals with the ML(MachineLearning) modules.And navigator is available in all the Windows,Linux and MacOS.The anaconda prompt has many number of IDE's which make the coding easier. The UI can also be implemented in python.

Standard Used: ISO/IEC 27001

Jupyter

It's like an open source web application that allows us to share and create the documents which contains the live code, equations, visualizations and narrative text. It can be used for data cleaning and transformation, numerical simulation, statistical modeling, data visualization, machine learning.

Standard Used: ISO/IEC 27001

Google Colab

Google Colab is a free playground for anyone wanting to explore Python in the cloud. No downloads, just open your browser and start coding! It provides access to powerful computing resources like GPUs and TPUs, making it ideal for machine learning, data analysis, or simply learning the ropes. You can even collaborate on projects with friends in real-time, sharing and editing notebooks like magic. While

resources are shared and temporary, it's perfect for quick experiments and learning bursts.

Standard Used: ISO/IEC 27001

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

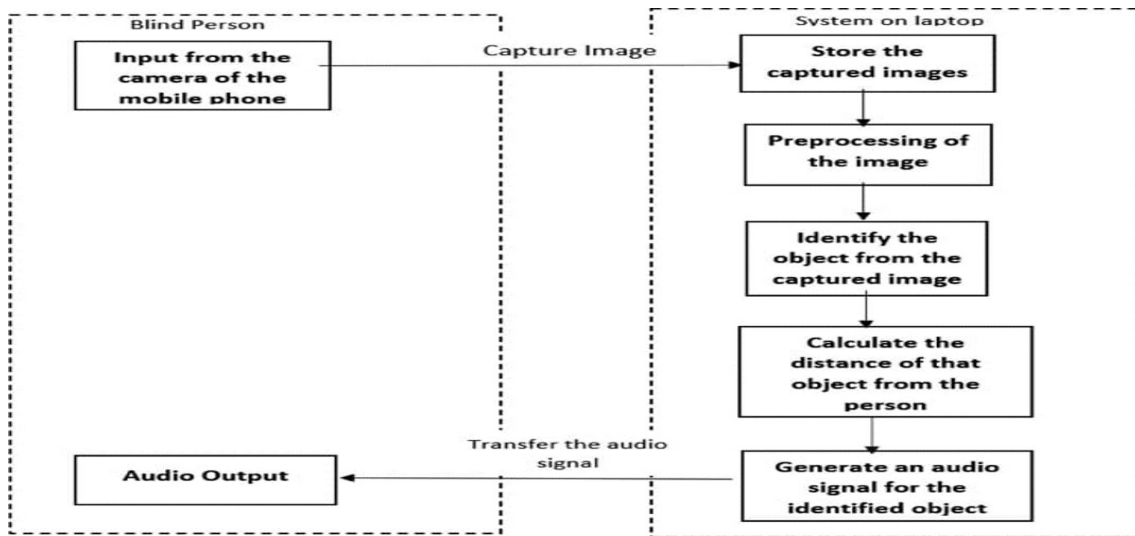


Figure 4.1: Architecture Diagram of Object Detection

Figure 4.1 depicts the architecture diagram for object detection for impaired individuals showcasing a multi-layered system. At its core is the "Detection Engine," utilizing deep learning algorithms for real-time object recognition. Input is received from sensors like cameras or LiDAR, processed by the engine, and output to the "Feedback Module." This module interprets the data and provides feedback through auditory, tactile, or visual interfaces. User preferences are stored in the "Settings Database," allowing customization. Optionally, a "Feedback Collection" component gathers user interactions for algorithm refinement. The system is scalable, with cloud integration for updates and data storage, ensuring continuous improvement and accessibility.

4.2 Design Phase

4.2.1 Data Flow Diagram

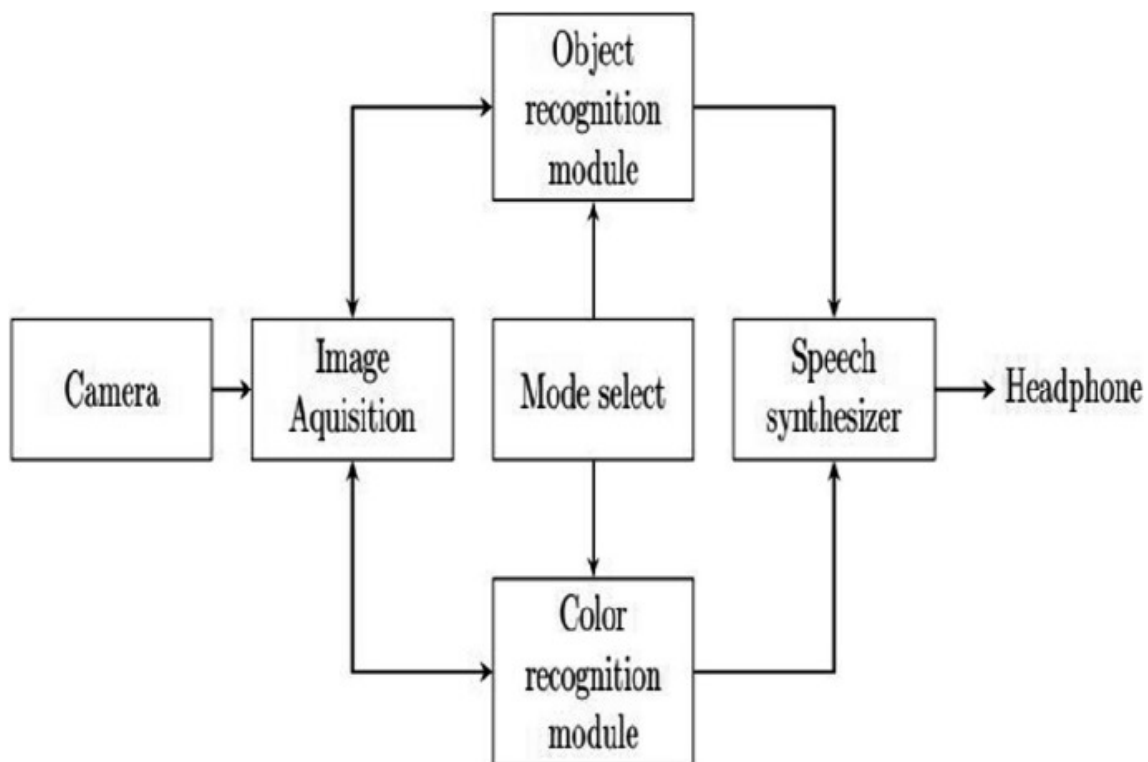


Figure 4.2: Data Flow Diagram of Object Detection

Figure 4.2 depicts the data flow diagram for object detection for impaired people illustrates the flow of information within the system. Input data, such as images or sensor readings, are captured from the environment and fed into the object detection algorithm. The algorithm processes the data and identifies objects present in the environment. Detected object information is then transmitted to the feedback module, where it is transformed into auditory, tactile, or visual cues for the user. User interaction data, such as settings adjustments or feedback, may also be captured and utilized to improve the object detection algorithm in future iterations.

Once objects are identified, the results are forwarded to the Feedback Generation Module, where the information is transformed into user-friendly output formats. Depending on user preferences and accessibility requirements, feedback may be delivered as auditory cues (via text-to-speech engines), tactile responses (through vibrations or Braille displays), or optional visual aids. The DFD also includes a User Interaction and Control Module, which captures user inputs such as setting adjustments, customization preferences, or system feedback.

4.2.2 Use Case Diagram

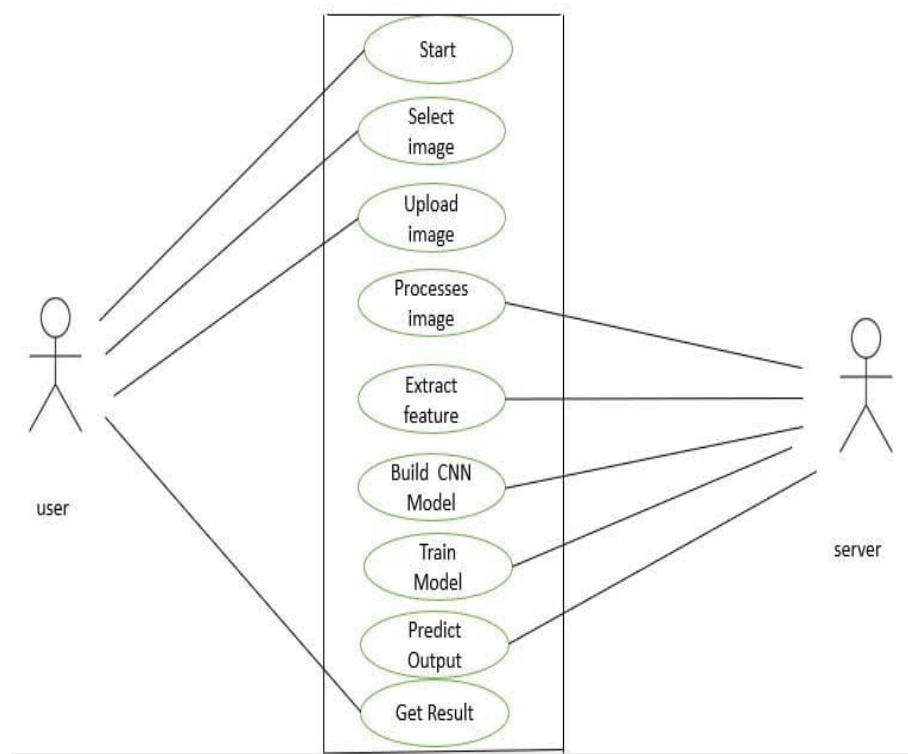


Figure 4.3: Use Case Diagram of Object Detection

Figure 4.3 depicts the use case diagram for object detection for impaired people illustrates interactions between the user and the system. The "Detect Objects" case captures input from sensors, processes it using object detection algorithms, and provides feedback on detected objects. "Provide Feedback" conveys information to the user through auditory, tactile, or visual cues. "Adjust Settings" allows customization, while "Access Help or Support" offers assistance features. "Update System" ensures the system remains current, and "Provide Training Data" optionally allows user feedback for algorithm improvement. This diagram showcases essential functionalities, including detection, feedback, customization, support, system maintenance, and user involvement in enhancing algorithm accuracy.

Additionally, the "Access Help or Support" use case enables users to receive troubleshooting assistance or guidance on using the system through voice commands or accessible help interfaces. The "Update System" ensures the software remains up to date with the latest enhancements, security patches, and model improvements. Meanwhile, the "Provide Training Data" use case allows users to contribute labeled images or feedback that can be used to retrain or fine-tune the object detection model, thereby improving its accuracy over time.

4.2.3 Class Diagram

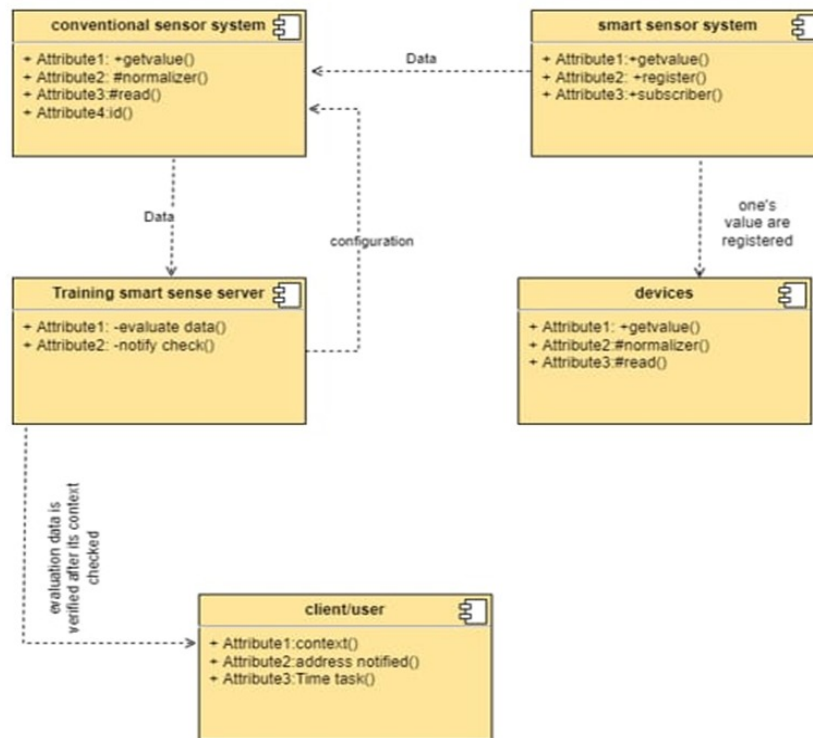


Figure 4.4: Class Diagram of Object Detection

Figure 4.4 presents the Class Diagram for the object detection system designed for visually impaired individuals, illustrating the core classes, their attributes, and the relationships that define the system's architecture. At the center of the design is the User class, which represents the visually impaired individual interacting with the system. This class maintains user-specific data such as identification, preferences, and interaction history. The DetectionSystem class encapsulates the primary functionalities related to image acquisition, preprocessing, object detection using CNN models, and result handling. It serves as the central controller coordinating the flow of data and execution of tasks.

Supporting this core logic are several interconnected classes. The DetectedObject class is responsible for storing structured data about each identified object, including labels, confidence scores, and location coordinates. The FeedbackModule class generates appropriate feedback based on the detected objects, translating data into auditory, tactile, or optional visual cues depending on user preferences. These preferences are managed by the Settings class, which allows the user to configure aspects such as feedback mode, voice type, or detection sensitivity.

4.2.4 Sequence Diagram

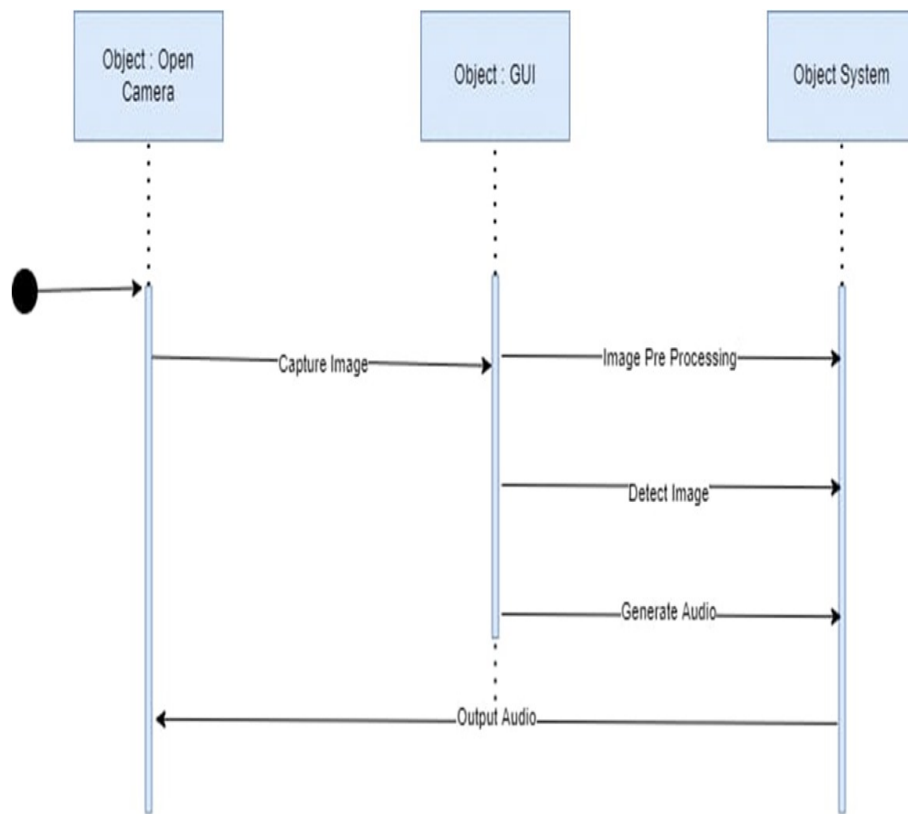


Figure 4.5: Sequence Diagram of Object Detection

Figure 4.5 illustrates the Sequence Diagram for the object detection system developed for visually impaired individuals, detailing the dynamic interactions and chronological flow of messages between system components and the user. The sequence begins when the User initiates a detection request, which triggers the DetectionSystem to activate the input module and capture environmental data through sensors or image files. This input is then forwarded to the ObjectDetectionEngine, where the Convolutional Neural Network (CNN) processes the data to identify and classify objects present in the scene.

Once object detection is completed, the recognized objects are sent to the FeedbackModule, which determines the appropriate output modality based on user preferences retrieved from the Settings module. The feedback—whether auditory via text-to-speech, tactile through vibrations, or visual—is then delivered back to the user to convey the detected information clearly and accessibly.

4.2.5 Collaboration diagram

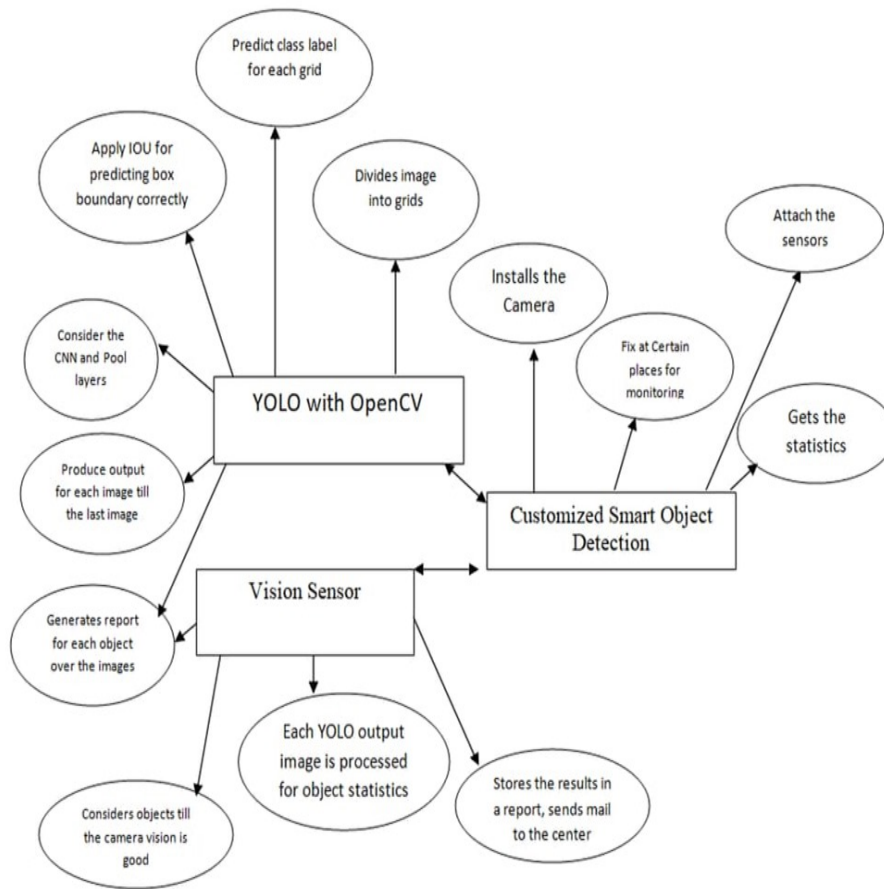


Figure 4.6: Collaboration diagram

Figure 4.6 depicts the Entity-Relationship (ER) Diagram for the object detection system tailored for visually impaired individuals, illustrating the key entities involved and the relationships between them. Central to the diagram is the User entity, representing the visually impaired individual interacting with the system. Each user can initiate one or more DetectionRequests, which are processed by the System entity. The system handles incoming data, performs object detection using CNN-based algorithms, and generates records of each DetectedObject, which are linked back to the detection sessions and user.

The System entity is also responsible for managing Settings on a per-user basis, allowing customization of preferences such as feedback type (auditory, tactile, or visual), language, and detection sensitivity. Optionally, the User may submit FeedbackData, contributing real-world corrections or responses that are stored and used to enhance the system's object detection algorithms over time.

4.2.6 Activity Diagram

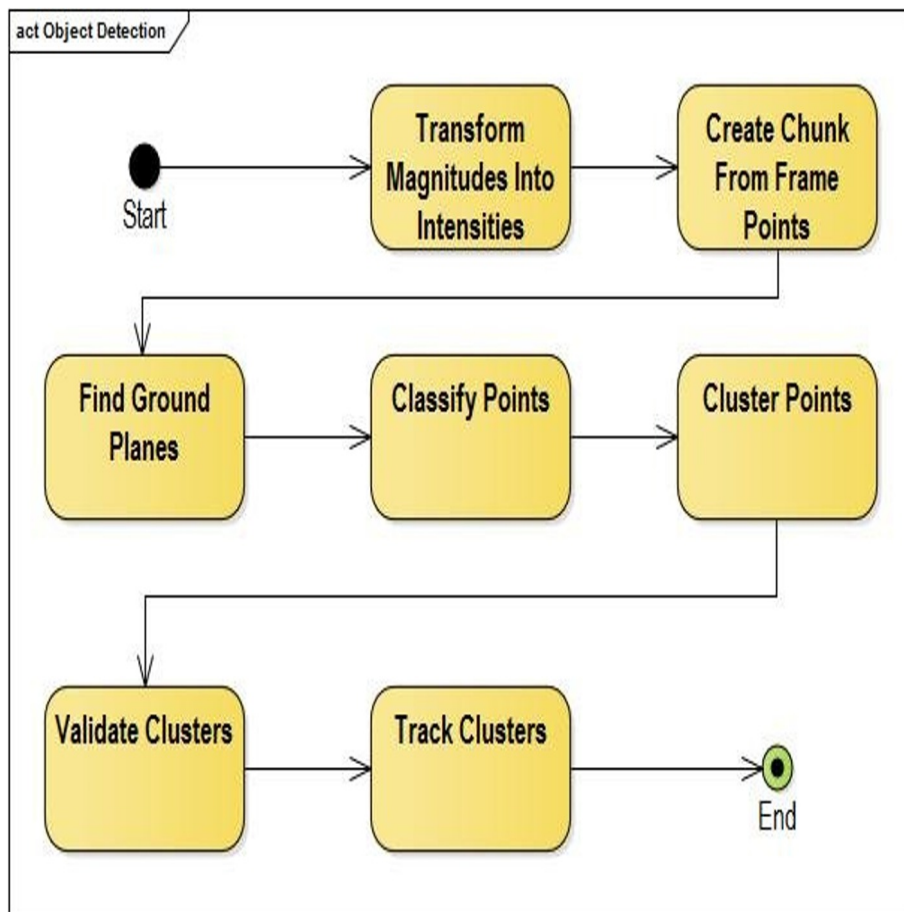


Figure 4.7: Activity Diagram of Object Detection

4.3 Algorithm & Pseudo Code

4.3.1 Convolutional Neural Network Algorithm

Convolutional Neural Network, particularly in combination with object detection algorithms, holds immense potential to assist impaired individuals in navigating their surroundings more independently. By leveraging CNNs, devices like smart glasses or smartphones equipped with cameras can recognize objects in real-time and provide auditory or tactile feedback to the user.

For example, a CNN-powered object detection system can identify common objects such as doors, chairs, or obstacles, and relay this information to the user through audio cues or vibrations. This allows visually impaired individuals to better navigate their environment, avoid obstacles, and locate desired objects. Training a CNN involves feeding it labeled data and optimizing its parameters to minimize the discrepancy between predicted and actual outputs. This process, known as backpropa-

gation, iteratively adjusts the weights of the network using optimization algorithms like stochastic gradient descent (SGD) or Adam. Through this iterative optimization process, the CNN learns to extract meaningful features from raw data and generalize its knowledge to unseen examples.

CNNs have achieved remarkable success in various computer vision tasks, including image classification, object detection, facial recognition, and image segmentation. Their ability to automatically learn hierarchical representations of features directly from raw data makes them incredibly powerful tools for analyzing and understanding visual information. Moreover, CNNs have been adapted and extended to other domains beyond computer vision speech recognition.

4.3.2 Pseudo Code

```
1 # Import necessary libraries
2 import numpy as np
3 import tensorflow as tf
4
5 # Define the CNN architecture
6 model = tf.keras.models.Sequential([
7     # Convolutional layers
8     tf.keras.layers.Conv2D(32, (3, 3), activation='relu', input_shape=(image_height, image_width, 3)
9     ),
10    tf.keras.layers.MaxPooling2D((2, 2)),
11    tf.keras.layers.Conv2D(64, (3, 3), activation='relu'),
12    tf.keras.layers.MaxPooling2D((2, 2)),
13    # Flatten layer to feed into fully connected layers
14    tf.keras.layers.Flatten(),
15    # Fully connected layers
16    tf.keras.layers.Dense(128, activation='relu'),
17    tf.keras.layers.Dense(num_classes, activation='softmax')
18 ])
19
20 # Compile the model
21 model.compile(optimizer='adam',
22               loss='sparse_categorical_crossentropy',
23               metrics=['accuracy'])
24
25 # Train the model
26 model.fit(train_images, train_labels, epochs=num_epochs, validation_data=(validation_images,
27                                   validation_labels))
28
29 # Evaluate the model
30 test_loss, test_acc = model.evaluate(test_images, test_labels)
31
32 # Use the model for object detection
```

4.4 Module Description

4.4.1 Deep Learning Frameworks

Deep learning frameworks have significantly advanced the development of assistive technologies aimed at improving the lives of visually impaired individuals, particularly in the domain of object detection. These frameworks provide the computational backbone for training and deploying models capable of identifying objects in real-time from image or video streams captured by cameras or other sensors. Several deep learning frameworks have emerged as key players in this field, each offering unique features and advantages. One prominent framework is TensorFlow, developed by Google Brain. TensorFlow provides a comprehensive ecosystem for building and deploying deep learning models, including tools for data preprocessing, model training, and deployment on various platforms. With its high-level APIs such as TensorFlow Object Detection API, developers can quickly build and train custom object detection models tailored to the needs of visually impaired users. These models can detect and classify objects in real-time, enabling applications such as navigation aids or object recognition systems.

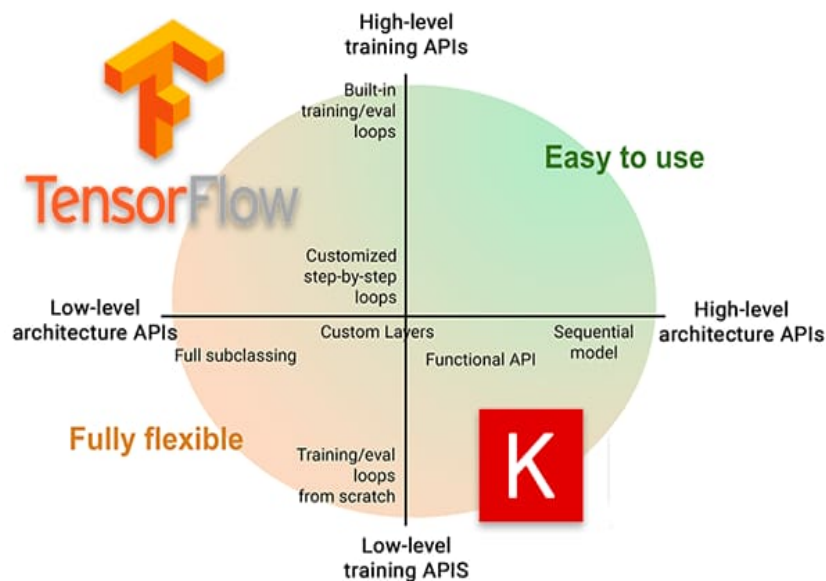


Figure 4.8: Deep Learning Frameworks

4.4.2 Pre-Trained Object Detection Models

Pre-trained object detection models serve as a cornerstone in developing assistive technologies for visually impaired individuals, offering efficient and accurate object recognition capabilities. These models have been trained on extensive datasets containing diverse objects and scenes, enabling them to identify objects in real-time with high precision. Leveraging pre-trained models expedites the development process by eliminating the need for extensive data collection and training, making it easier to deploy object detection systems tailored to the needs of visually impaired users. Faster CNN stands out as a popular choice in the realm of pre-trained object detection models. Renowned for its speed and accuracy, Faster CNN combines a region proposal network (RPN) with a convolutional neural network (CNN) backbone to detect objects across various scales and aspect ratios. Pre-trained on datasets like COCO (Common Objects in Context), Faster CNN can discern a wide array of objects, making it a versatile solution for assistive applications. It offers another compelling option, prioritizing real-time performance. divides the input image into a grid and directly predicts bounding boxes and class probabilities for each grid cell. Its balance between speed and accuracy makes it ideal for applications requiring rapid ob

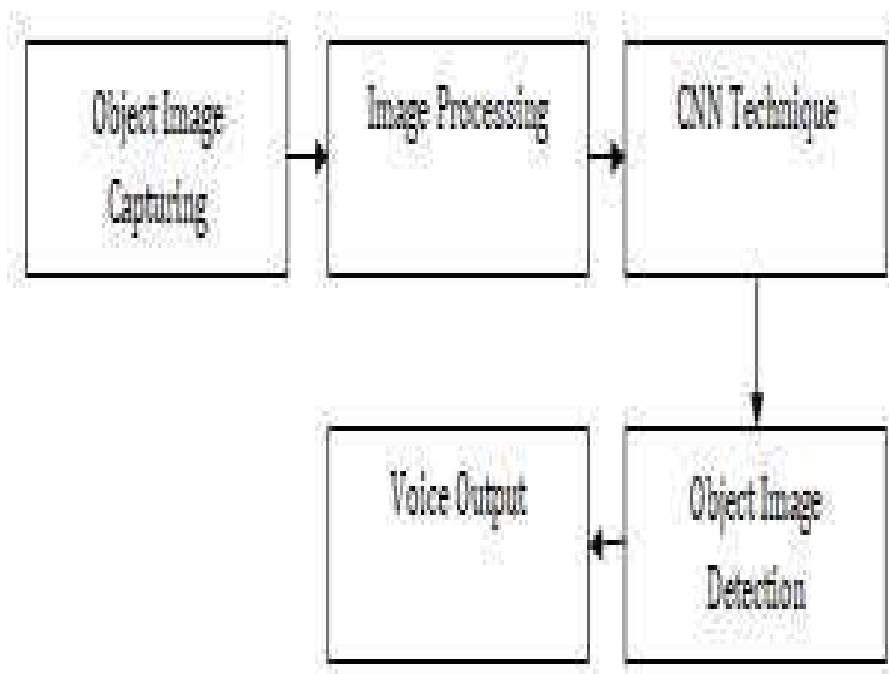


Figure 4.9: Pre-Trained Object Detection

4.4.3 Setup Speech and Audio Processing

Setting up speech and audio processing for object detection to aid visually impaired individuals involves integrating audio feedback and speech recognition with object detection systems to provide real-time assistance. The process typically begins with capturing audio inputs from microphones or other audio sensors, which are then processed to extract relevant features. Speech recognition algorithms are employed to convert spoken commands or queries into text, enabling users to interact with the system verbally. Simultaneously, object detection models, such as those mentioned previously, analyze visual inputs from cameras or image sensors to identify objects in the user's environment. These models can be pre-trained on extensive datasets to recognize a wide range of objects with high accuracy. Once an object is detected, relevant information about its identity, location, and characteristics is extracted from the visual data. The audio feedback component of the system converts this extracted information into speech or audio cues that are relayed to the user through headphones, earpieces, or speakers. This feedback informs the user about the presence and properties of detected objects in their surroundings. For example, upon detecting a "chair," the system may audibly announce the object's name and location relative to the user, such as "Chair located three meters ahead."

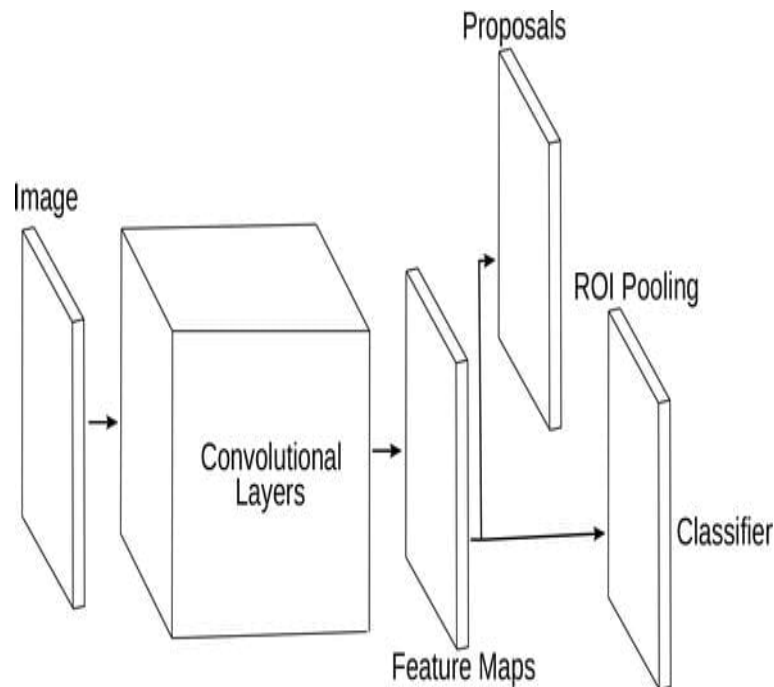


Figure 4.10: Setup Speech and Audio Processing

4.5 Steps to execute/run/implement the project

4.5.1 Research and Understand User Needs

Gain insights into the specific challenges faced by visually impaired individuals regarding object recognition. Understand their requirements, preferences, and constraints.

4.5.2 Preprocess Data

Prepare the dataset by resizing images, normalizing pixel values, and splitting it into training, validation, and test sets.

4.5.3 Implement Accessibility Features

Integrate accessibility features into the system, such as audio feedback, text-to-speech capabilities, or tactile interfaces, to assist visually impaired users in interacting with the detected objects.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

Input testing for object detection for impaired individuals involves assessing the system's accuracy in detecting common objects, performance in complex environments and adverse conditions, responsiveness to user interactions, effectiveness of feedback modalities, customization options, and stability over prolonged use, ensuring reliable and intuitive functionality for visually impaired users.

5.1.2 Output Design

Output testing for object detection for impaired individuals verifies the system's ability to provide accurate and timely feedback through auditory alerts, tactile vibrations, or visual cues. It evaluates the clarity, reliability, and effectiveness of feedback in conveying object information to visually impaired users for safe navigation and interaction with their environment.

5.2 Testing

Testing for object detection for impaired individuals involves comprehensive input testing to ensure accurate detection of common objects in various environments and adverse conditions. Output testing verifies effective feedback provision through auditory, tactile, or visual cues, ensuring reliable and intuitive functionality for visually impaired users.

5.2.1 Unit Testing

```
1 # Checkbox to toggle voice output
2 stop_voice = st.checkbox("Toggle Voice Output")
```



```

3
4 # Speaking the prediction
5 if not stop_voice:
6     try:
7         engine.say(prediction_text)
8         engine.runAndWait()
9     except RuntimeError:
10         pass # Handle RuntimeError gracefully

```

Unit testing for object detection in accessibility-focused systems ensures accurate detection of objects crucial for visually impaired users. It verifies individual components of the detection process, including preprocessing, model inference, and output formatting, to guarantee reliability and precision. Tests assess various scenarios, such as different lighting conditions and object orientations, ensuring consistent performance. Additionally, accessibility features like audio feedback are validated for usability and effectiveness, enhancing the overall user experience.

5.2.2 System Testing

```

1 import tensorflow as tf
2 from tensorflow.keras.models import load_model
3 import streamlit as st
4 import numpy as np
5 import pyttsx3
6
7 # Initialize the text-to-speech engine
8 engine = pyttsx3.init()
9 voices = engine.getProperty('voices')
10 engine.setProperty('voice', voices[1].id) # Select a female voice, adjust index as needed
11
12 st.header('Image Classification Model')
13
14 try:
15     # Attempt to load model
16     model_path = '\\Users\\pudur\\Downloads\\Image_classification\\Image_classify.keras'
17     model = load_model(model_path)
18 except Exception as e:
19     st.error(f"Error loading model: {e}")
20     st.stop()
21
22 data_cat = ['apple', 'banana', 'beetroot', 'bell pepper', 'cabbage', 'capsicum', 'carrot', 'cauliflower',

```

```

23         'chilli pepper', 'corn', 'cucumber', 'eggplant', 'garlic', 'ginger', 'grapes', 'jalepeno
24         ', 'kiwi', 'lemon',
25         'lettuce', 'mango', 'onion', 'orange', 'paprika', 'pear', 'peas', 'pineapple', '
26         pomegranate', 'potato',
27         'raddish', 'soy beans', 'spinach', 'sweetcorn', 'sweetpotato', 'tomato', 'turnip', '
28         watermelon']
29
30 img_height = 180
31 img_width = 180
32
33 image = 'corn1.jpg'
34 uploaded_file = st.file_uploader("Upload an image", type=["jpg", "jpeg", "png"])
35 if uploaded_file is not None:
36     image = uploaded_file.name
37
38 image_load = tf.keras.utils.load_img(image, target_size=(img_height, img_width))
39 img_arr = tf.keras.utils.img_to_array(image_load)
40 img_bat = np.expand_dims(img_arr, axis=0)
41
42 predict = model.predict(img_bat)
43 score = tf.nn.softmax(predict)
44
45 st.image(image, width=200)
46
47 prediction_text = data_cat[np.argmax(score)]
48 accuracy_text = ('With ') + str(round((np.max(score) * 100), 3)) + ' % accuracy '

```

System testing for object detection systems catering to visually impaired individuals involves evaluating the entire system's functionality and performance. This comprehensive testing assesses the system's ability to accurately detect objects in diverse environments and lighting conditions, ensuring reliability in real-world scenarios. Additionally, it verifies the effectiveness of accessibility features such as audio feedback and text-to-speech capabilities in assisting visually impaired users. System testing validates the system's usability, accuracy, and robustness, ultimately enhancing accessibility for impaired individuals.

5.2.3 Input

✓ Last week

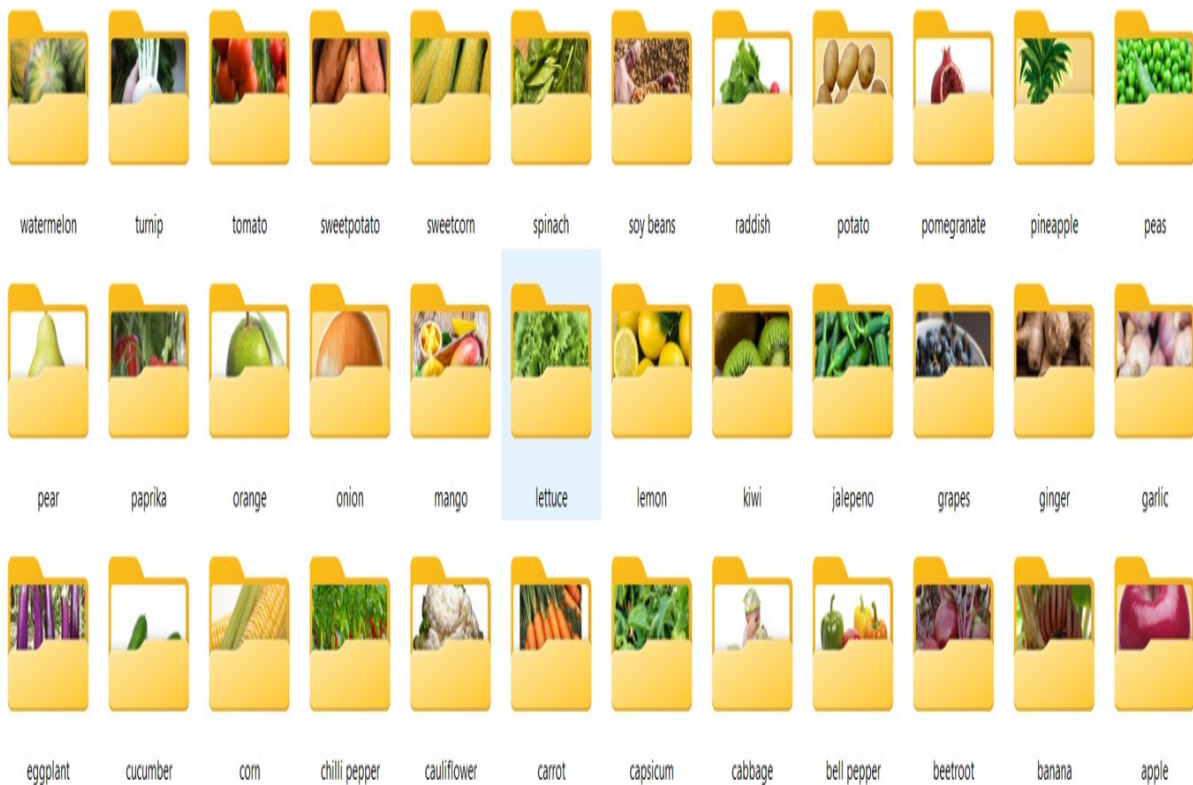


Figure 5.1: Test Image of Object Detection

The provided image showcases a well-organized dataset directory, where each folder represents a distinct class of fruits or vegetables such as apple, banana, tomato, spinach, lettuce, mango, and many others. This structure is typically used in image classification tasks, particularly with convolutional neural networks (CNNs), to facilitate supervised learning. Each folder contains multiple images of the corresponding item, making it suitable for training, validation, and testing of deep learning models. The visual thumbnails on the folder icons offer a quick preview of the images inside, helping to verify the dataset content at a glance. This type of dataset organization is commonly utilized with frameworks like TensorFlow or PyTorch using directory-based data loaders.

5.2.4 Testing Strategies

5.2.5 Performance Evaluation

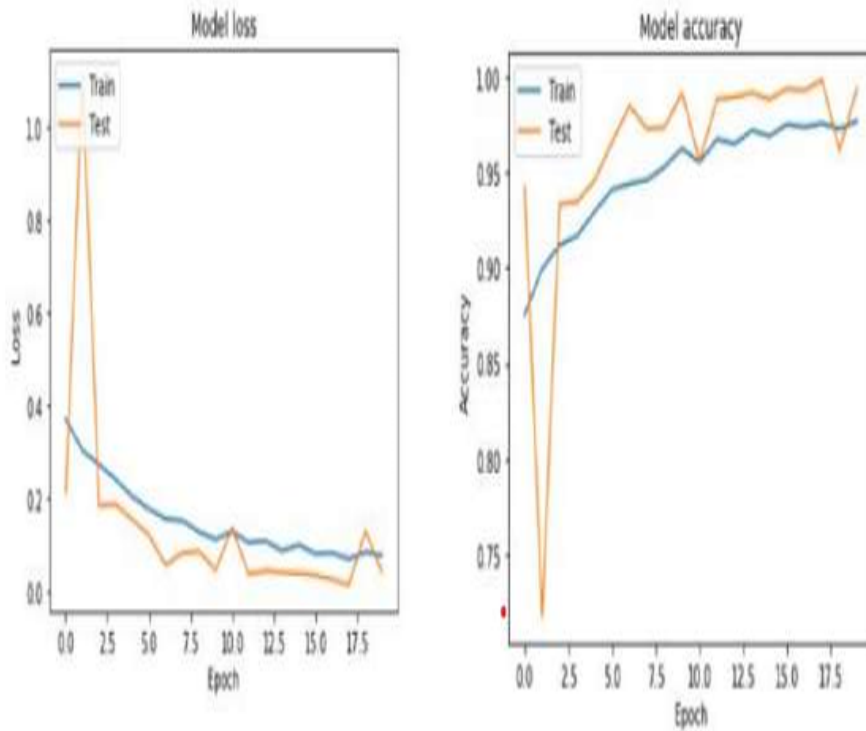


Figure 5.2: Test Image

The above figure illustrates the training and validation performance of the Convolutional Neural Network (CNN) model across 25 epochs. The left panel represents the Accuracy, while the right panel depicts the Loss trend.

In the Accuracy plot, both training and validation accuracies exhibit a steep rise within the initial 5 epochs, indicating rapid learning. The training accuracy approaches nearly 100, while the validation accuracy stabilizes around 95, demonstrating a strong generalization capability of the model. The slight divergence between the two curves after epoch 15 may indicate a minor overfitting tendency, though not significantly impactful. In the Loss plot, both curves show a sharp decline during the early training stages, with training loss eventually dropping below 0.1. The validation loss levels off slightly higher than the training loss, hovering around 0.5, which is acceptable given the model complexity and input noise.

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The efficiency of a proposed object detection system for impaired people is critical for its practical adoption and usability. The system's efficiency encompasses several key aspects. Firstly, the accuracy and reliability of the object detection algorithm are paramount. The system must be able to detect and classify objects in real-time with high precision, especially in dynamic and cluttered environments where visually impaired individuals navigate daily. Utilizing state-of-the-art deep learning models like YOLO (You Only Look Once) or SSD (Single Shot MultiBox Detector) can enhance accuracy while maintaining efficient inference speeds suitable for real-time applications on resource-constrained devices.

Secondly, the efficiency of the system involves optimizing computational resources. Object detection algorithms should be optimized to run efficiently on the target hardware, such as smartphones or wearable devices, to minimize processing time and conserve battery life. Techniques like model quantization, pruning, and hardware acceleration (e.g., using GPU or specialized neural processing units) can significantly improve computational efficiency without compromising accuracy.

Additionally, the system's efficiency extends to the user interface and interaction design. The user interface should be intuitive and accessible, providing clear and timely feedback to visually impaired users about detected objects. Audio-based feedback, tactile vibrations, or voice commands can enhance the system's usability, enabling seamless integration into daily life activities.

Moreover, the efficiency of the proposed system involves robustness and adaptability. The system should be capable of handling various environmental conditions, such as changes in lighting or occlusions, to maintain consistent performance. Continuous improvement through adaptive learning techniques, such as online fine-tuning of models based on user feedback, can enhance the system's adaptability and responsiveness over time.

6.2 Comparison of Existing and Proposed System

Existing System: In traditional systems for plant disease detection, the process is either manual or based on simple image processing techniques. These systems rely heavily on predefined features like color, texture, and shape for classification. However, such approaches lack the intelligence to generalize across diverse plant species and various stages of disease progression. Manual inspection requires expert knowledge and is time-consuming, especially in large-scale farming. In most cases, there is no support for real-time feedback, and the outputs are only visual, which limits accessibility, especially for visually impaired users. The lack of automation and adaptability in these existing systems often leads to inconsistent results, lower accuracy, and delayed disease control, which can significantly impact crop yield and increase financial losses for farmers.

Proposed System: The proposed system addresses the limitations of the existing system by utilizing a Convolutional Neural Network (CNN) for intelligent, automated, and accurate plant disease detection. CNNs have the ability to learn from large datasets and extract deep visual features, making them highly effective for image classification tasks. The system not only processes and identifies diseases from plant images with high precision but also includes a voice output feature, allowing the results to be audibly announced. This enhances accessibility for users with visual impairments and those working in environments where screen interaction is not practical. Additionally, the proposed system supports image file input, enabling batch processing and easy integration into existing agricultural workflows. By combining deep learning with voice-enabled feedback, the system offers a scalable, efficient, and inclusive solution that minimizes human error and supports timely intervention for disease control.

6.3 Comparative Analysis-Table

Feature	Existing Systems	Proposed System
Disease Detection Method	Manual inspection or basic image processing techniques	Advanced Convolutional Neural Network (CNN) for accurate detection and classification
Accuracy and Reliability	Often subjective and less accurate	High accuracy through deep learning trained on large datasets
User Accessibility	Primarily visual interface only	Voice output makes it accessible for visually impaired and less tech-savvy users
Input Method	Real-time camera only in most systems	Supports both real-time camera input and batch image files
Processing Time	Slower and may require expert interpretation	Fast, automated detection using pre-trained model
Scalability	Limited scalability and usability in large farms	Easily scalable and can be integrated with agricultural monitoring systems
Internet Dependency	Many tools rely on cloud-based processing	Can be configured to run offline using local models and TTS engines
Remedy Suggestions	Often requires manual lookup or expert help	Can be extended to provide immediate suggestions for treatment of identified diseases
Cost of Operation	High, especially if expert diagnosis or equipment is needed	Low-cost solution suitable for small and marginal farmers
Target Users	Agricultural experts or trained personnel	Designed to assist common farmers, especially those with limited access to expert support

Table 6.1: Comparison of Existing and Proposed System

The table presented above outlines a comparative analysis between traditional (existing) systems used for plant disease detection and the proposed intelligent solution that integrates Convolutional Neural Networks (CNN) with voice output. Existing systems often rely on manual inspection or rudimentary image processing techniques, which can lead to inaccurate or subjective results due to human error or limited feature extraction capabilities. In contrast, the proposed system employs a robust CNN model trained on a large dataset, ensuring significantly higher accuracy and reliable classification of diseases.

6.4 Comparative Analysis-Graphical Representation and Discussion

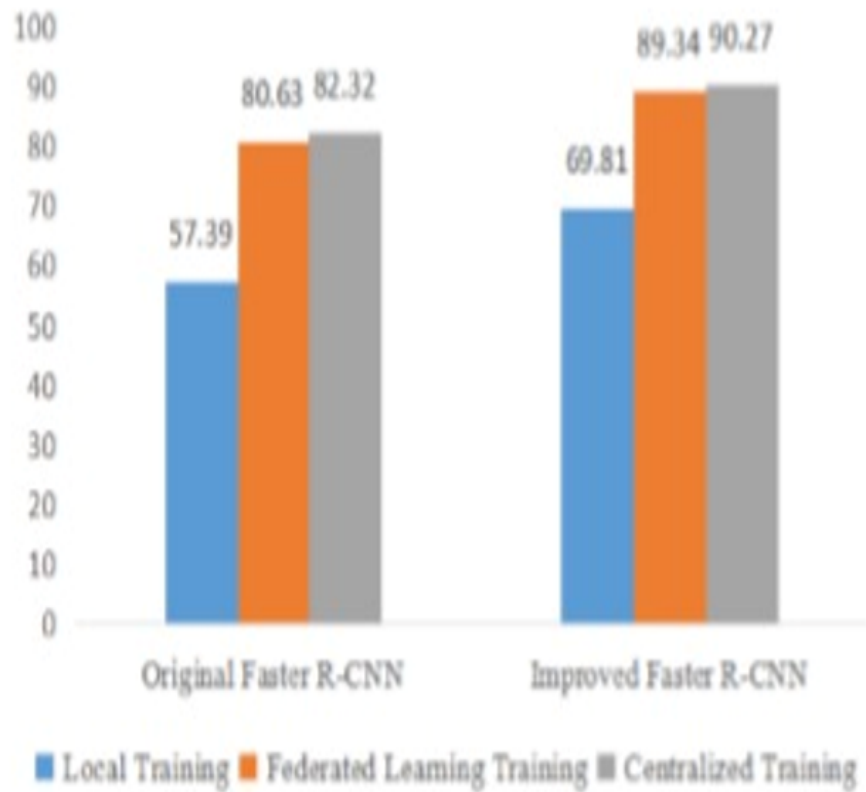


Figure 6.1: **Graphical Representation**

The graphical representations included in this project provide visual insights into the performance and efficiency of the proposed system compared to existing methodologies. These graphs serve to illustrate key metrics such as model accuracy, processing time, classification results, and user accessibility enhancements. For instance, bar charts or line graphs may be used to demonstrate the comparative mean Average Precision (mAP) achieved by different models, highlighting the superior performance of the proposed CNN-based approach.

Chapter 7

INDUSTRY DETAILS

7.1 Cognizant Technology Solutions

7.1.1 (19/12/2024 - 19/04/2025)

7.1.2 4Months

7.1.3 Manyata Tech Park,Nagavara,Bengaluru,Karnataka

7.2 Internship offer letter



09-Dec-2024

Candidate ID: 32688744

Bolla Leela Krishna Mohan
B.Tech Information Technology
Vel Tech Rangarajan Dr. Sagunthala R & D Institute of Science and Technology, Chennai

Dear Bolla Leela Krishna Mohan,

Further to our Letter of Intent for the position of Programmer Analyst Trainee / Programmer Analyst aligned to the hiring category, we are pleased to offer you an internship with us at Cognizant office for a period of 3 to 6 months. Your internship on-boarding will be scheduled based on your availability factoring your college exam schedule, availability of your Provisional Certificate and our business requirements.

During this period, you will be provided with a stipend of INR 12,000 per month equated to the planned duration of the Internship curriculum and will be paid only subject to successful completion of milestones as defined in the curriculum, prior to the monthly stipend processing window, for a given month based on your performance and attendance.

Though Cognizant Internship is a skilling program aimed at enhancing technical acumen, it does not guarantee employment and there is no employer - employee relationship during the course of this internship program. However, the successful completion of Internship will form a critical part of your eligibility for employment with Cognizant if an opportunity arises in future.

You will be provided a learning curriculum as per the skill track assigned to you. The learning design would expect you to drive your learning through hands on exercise and project work. There will also be series of webinars, quizzes, Subject Matter Expertise (SME) interactions, mentor connects, code challenges, assessments etc. to accelerate your learning. The outcome during Internship would be monitored through formal evaluations.

Prior to joining on the rolls of Cognizant, you must have successfully completed the prescribed Internship program. In the event of unsatisfactory performance during the Internship or non-completion of the Internship, no Internship Completion Certificate shall be issued by Cognizant. Cognizant reserves rights at its sole discretion to revoke its Letter of Intent.

Section A: Terms and Conditions:

1. The Internship timings would be for 10 hours per day from Monday through Friday aligned to the working timings followed in Cognizant which based on the need could also be operated on a shift model. Attendance is mandatory on all the days to stay active in the Internship Program. The Internship Offer would be cancelled if the mandatory requirement of minimum 85% attendance at office is not met in a month.

2. Interns are covered under Cognizant's calendar holidays of the respective location of internship, and you would need to adhere with minimum attendance requirements. Prior approvals are must towards any unavoidable leave or business requirements during the program and the internship would be cancelled if leaves are availed without prior approvals.

3. You would be required to ensure timely completion and submission of assignments, project work and preparation required prior to the sessions failing which your internship would be cancelled.

4. The skill track mapped could change at the start or mid-way or even later during the program depending on business needs and changes and you would be required to be flexible for this change failing which your internship would be cancelled.

Payment will be done for the prescribed Internship Curriculum period only and no additional payment will be done for any delay in completion. Attendance and successful completion of Milestone(s) are the eligible factors for processing stipend payment and tenure spent will not guarantee your monthly stipend payment.

Regd Office: 115/535, Old Mahabalipuram Road, Okkiam Thoraiakkam, Chennai - 600 097

7.3 Internship offer letter



6. There would be zero tolerance to plagiarisms and misconduct during the internship. Adherence to Cognizant Internship policies and guidelines is mandatory and any breach or incident reported will lead to immediate cancellation of Internship without any notice. You would be required to complete Cognizant mandatory trainings such as Code of Conduct and Acceptable Use Policy (AUP) within the given timelines.

7. During the course of your Internship and at all times, you shall be governed by Cognizant's Social Media Policy and shall refrain from posting malicious, libelous, defamatory, false, obscene, political, anti-social, abusive, and threatening messages/statements or disparaging the Company, clients, associates, competitors, or suppliers or any third parties, irrespective of whether any such statements are likely to cause damage to any such entity or person. Any breach of this section would lead to immediate cancellation of the Internship.

8. Cognizant reserves rights regarding IT infra as applicable and access to information and material of Cognizant during the internship period and may modify or amend the Cognizant GenC program terms and conditions from time to time.

9. It is hereby clarified that participation in this Internship shall not constitute you to be an employee of Cognizant nor obligates Cognizant for any purpose whatsoever. The scope of this Internship does not include any supervisory responsibilities and that there is no agency, fiduciary or employer-employee relationship intended or created by reason of this document.

10. Cognizant holds all rights to cancel this Internship Offer due to non-conformance of performance benchmark or moral code of conduct or in case of you failing to participate in the Internship within the given date/timeline or for such other any reasons upon providing written communication of the same to you. Upon such cancellation of this Internship Offer, your access and participation in the Internship shall stand cancelled.

11. At the time of your reporting for the internship, you will be required to sign a Non – Disclosure Agreement with the company. During the course of your internship and after completion of the same, you are required to maintain strictest confidentiality with respect to company proprietary or products that you access or come into contact with, during your project as an Intern, at all times as per our Policy. Use of company proprietary information or products shall not be made without prior permission from the concerned authority. Any breach of information security will be dealt as per Company Policy.

12. After successful completion of your internship if there is a business demand which expects you to get enabled on a different skill, you would be onboarded as full-time employee (FTE) and deployed into another formal training based on business demand to a specific skill track which will be used as basis towards your allocation to projects/roles. Successful completion of this training is mandatory to continue as an FTE with Cognizant.

13. This offer from Cognizant shall be active and **valid for only 3 calendar days** and hence you are expected to accept or decline the offer through the company's online portal within the said time-period of 3 calendar days and you will also be required to submit the mandatory documents at least **7 days** before your Internship Onboarding Date as part of your Pre-joining & Background Verification (BGV) process. In case you don't comply to the above timelines or if the background verification checks reveal unfavorable results at any time, this Offer shall stand withdrawn and will be considered as cancelled. Any official written extension to the offer validity and the above-mentioned timelines will be at the sole discretion of Cognizant.

14. For avoidance of doubt, it is herewith stated that the Internship shall stand cancelled on the below scenarios as well:

- a. In the event of you accepting this Internship Offer but not joining into the Internship on the specified date and at the specified location of on-boarding.
- b. In the event of you not accepting this Internship Offer or failing to communicate acceptance within 3 calendar days as stated above, the opportunity to do internship will be cancelled.
- c. For such other operational, regulatory reasons including breach of terms herein.

Thereupon, your access shall also stand revoked, and Cognizant shall not be obligated to extend nor be liable for any claims due to cancellation of this Internship Offer.

On any of the above-mentioned scenarios (Refer to **Section A: Terms and Conditions**), if your Internship Offer has been cancelled then your Letter of Intent would also be revoked at the sole discretion of Cognizant.

Below are the **mandatory documents** to be submitted as part of your **Background Verification**:

- Your Pan Card
- Letter of Authorization (LOA) which should be downloaded from the BGV application hand signed with your name and date and re-uploaded back to the application

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7.4 Internship offer letter



Below are the **mandatory documents** to be submitted as part of your **Pre-joining formalities**:

- 2 Passport sized Photographs preferably with a Grey / White background
- Personal individual bank account from a nationalized bank for processing stipend

In case of additional queries or concerns, you can raise a query at

<https://campus2Cognizant.cognizant.com>

We wish you good luck.

Yours sincerely,

For Cognizant Technology Solutions India Pvt. Ltd.,



Maya Sreekumar
Vice President - Human Resources

I have read the offer, understood and accept the above mentioned terms and conditions.

Signature:

Date:

Regd Office: 115/535, Old Mahabalipuram Road, Okkiam Thoraipakkam, Chennai - 600 097

7.5 Internship Completion certificate

Chapter 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1 Summary

In conclusion, object detection technology represents a transformative solution for enhancing accessibility and independence among visually impaired individuals. Through the integration of advanced computer vision algorithms and assistive devices, object detection systems empower users by providing real-time information about their surroundings. This technology plays a crucial role in improving spatial awareness, facilitating safer navigation, and promoting social inclusion for individuals with visual impairments.

The development and adoption of object detection systems for impaired people signify a significant step towards creating more inclusive and equitable societies. By leveraging state-of-the-art machine learning models and accessible user interfaces, these systems cater to the specific needs and challenges faced by visually impaired individuals, allowing them to navigate environments with greater confidence and autonomy. Moreover, the continuous evolution of object detection technology holds promise for further advancements in assistive technologies, offering enhanced functionalities and usability over time.

Looking ahead, it is essential to continue advancing research and development efforts in this field while ensuring adherence to accessibility standards, data privacy regulations, and ethical guidelines. Collaboration among researchers, developers, policymakers, and advocacy groups is crucial to promote the responsible and inclusive deployment of object detection solutions for impaired individuals. Ultimately, by harnessing the potential of object detection technology, we can empower visually impaired individuals to overcome barriers, participate more actively in daily activities, and experience greater independence and quality of life.

8.2 Limitations

While the proposed system offers significant advancements in plant disease detection through the use of Convolutional Neural Networks and voice output, it still has certain limitations that affect its full potential. One of the primary limitations is the dependency on high-quality and well-labeled datasets. The performance of the CNN model is directly influenced by the quality and variety of the images used during training. If the dataset does not contain sufficient examples of different diseases or plant types, the model may struggle to accurately classify unseen data in real-world scenarios. Furthermore, lighting conditions, background noise in images, and image resolution may affect detection accuracy. This restricts the system's ability to generalize across diverse agricultural environments, especially in regions where ideal image capture conditions are not feasible.

Another limitation lies in the voice output functionality. While it enhances accessibility, especially for visually impaired users, it may not be effective in noisy agricultural environments or for users speaking different languages or dialects. The current implementation may only support limited languages and simple sentence structures, which could restrict its usability across multicultural or multilingual user bases. Additionally, the system currently processes images sequentially and may not be optimized for real-time performance in large-scale field applications. This makes it less practical for rapid disease monitoring across vast farmlands without further hardware or software optimization. Future work should address these issues by incorporating multilingual support, noise-canceling features in audio output, and real-time edge processing capabilities.

8.3 Future Enhancements

- 1. Improved Object Recognition Algorithms:** Continuously refining and optimizing object recognition algorithms can lead to more accurate and efficient detection of objects in various environments.
- 2. Multi-Sensory Integration:** Integrating multiple sensory inputs such as vision, sound, and touch can provide a more comprehensive understanding of the surroundings. For example, combining visual object detection with audio cues or haptic feedback can enhance accessibility for visually impaired individuals.
- 3. Real-Time Feedback Systems:** Developing real-time feedback systems that

provide immediate information about detected objects can help impaired individuals navigate their surroundings more effectively. This could include auditory alerts or vibrations to indicate the presence and location of objects.

4. Customizable Interfaces: Designing customizable interfaces that allow users to tailor the object detection system according to their specific needs and preferences can improve usability and accessibility.

5. Integration with Wearable Technology: Integrating object detection technology into wearable devices such as smart glasses or haptic vests can provide a seamless and hands-free experience for impaired individuals, allowing them to receive real-time object detection information directly through their wearable devices.

6. Machine Learning for Personalization: Leveraging machine learning algorithms to personalize the object detection system based on individual preferences, behaviors, and environments can enhance its effectiveness and usability over time.

7. Cloud-Based Object Recognition: Utilizing cloud-based object recognition services can provide access to a vast database of object models and enable real-time updates and improvements to the detection algorithms.

8. Collaboration with Accessibility Experts: Collaborating with accessibility experts and individuals with disabilities throughout the development process can ensure that the object detection system meets the specific needs and requirements of impaired users.

9. Privacy and Security Considerations: Incorporating robust privacy and security measures to protect the sensitive data collected by the object detection system is crucial to ensure the trust and adoption of the technology.

Chapter 9

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

9.1 Alignment with SDGs

The proposed project aligns strongly with the United Nations' Sustainable Development Goals by promoting technological innovation in agriculture, increasing accessibility, and supporting inclusive development. By leveraging Convolutional Neural Networks for accurate plant disease detection and integrating voice output for accessibility, the system addresses both agricultural efficiency and social inclusivity. This contributes to sustainable farming practices by minimizing crop loss, improving food production, and empowering farmers, especially those in under-resourced or rural areas.

Furthermore, the project's inclusive design—with voice support—ensures that individuals with visual impairments or limited literacy can also benefit, thereby reducing inequalities in access to agricultural tools and knowledge. The system's ability to work offline or in low-resource environments enhances its relevance in developing regions, making it a valuable tool in achieving sustainable development in agriculture and technology access.

9.2 Relevance of the Project to Specific SDG

SDG 2 – Zero Hunger: The system directly contributes to SDG 2 by supporting early detection of crop diseases, thereby reducing crop failure and improving food security. Timely diagnosis ensures better yields and helps in maintaining a stable food supply.

SDG 3 – Good Health and Well-Being: By reducing the spread of plant diseases and ensuring the health of crops, the system indirectly supports public health. Healthy crops mean fewer chances of pesticide overuse and safer food production.

SDG 9 – Industry, Innovation, and Infrastructure: The integration of AI and voice technology into agriculture fosters innovation and contributes to building resilient infrastructure for smart farming. This represents a step forward in modernizing traditional farming techniques.

SDG 10 – Reduced Inequalities: The inclusion of voice output functionality supports users with visual impairments or literacy barriers, promoting equality in access to technological tools and information.

SDG 12 – Responsible Consumption and Production: By helping farmers identify diseases early, the system promotes efficient use of agricultural inputs such as pesticides and fertilizers, reducing waste and environmental damage.

9.3 Potential Social and Environmental Impact

The proposed system holds substantial potential for positive social impact, particularly in rural and agricultural communities where access to expert guid-

ance and technological support is limited. By empowering farmers with an accessible, AI-driven tool for early detection of plant diseases, the system contributes to improving their livelihood and economic resilience. The voice output feature adds inclusivity by enabling visually impaired individuals and those with limited literacy to effectively interact with the system, thereby promoting social equity and digital inclusion in the agricultural sector. This can lead to greater independence and confidence among marginalized groups, contributing to inclusive growth and community development.

From an environmental standpoint, the early and accurate identification of plant diseases reduces the overuse and misuse of chemical pesticides and fertilizers. This not only minimizes soil and water pollution but also protects beneficial insects and biodiversity in farming ecosystems. By encouraging responsible agricultural practices, the system helps preserve natural resources and supports sustainable land use. Additionally, reducing crop failure through timely intervention can lower the carbon footprint associated with replanting and wasted resources. Overall, the project contributes to building a more sustainable and environmentally-conscious agricultural future, aligning technology use with ecological stewardship.

Chapter 10

PLAGIARISM REPORT

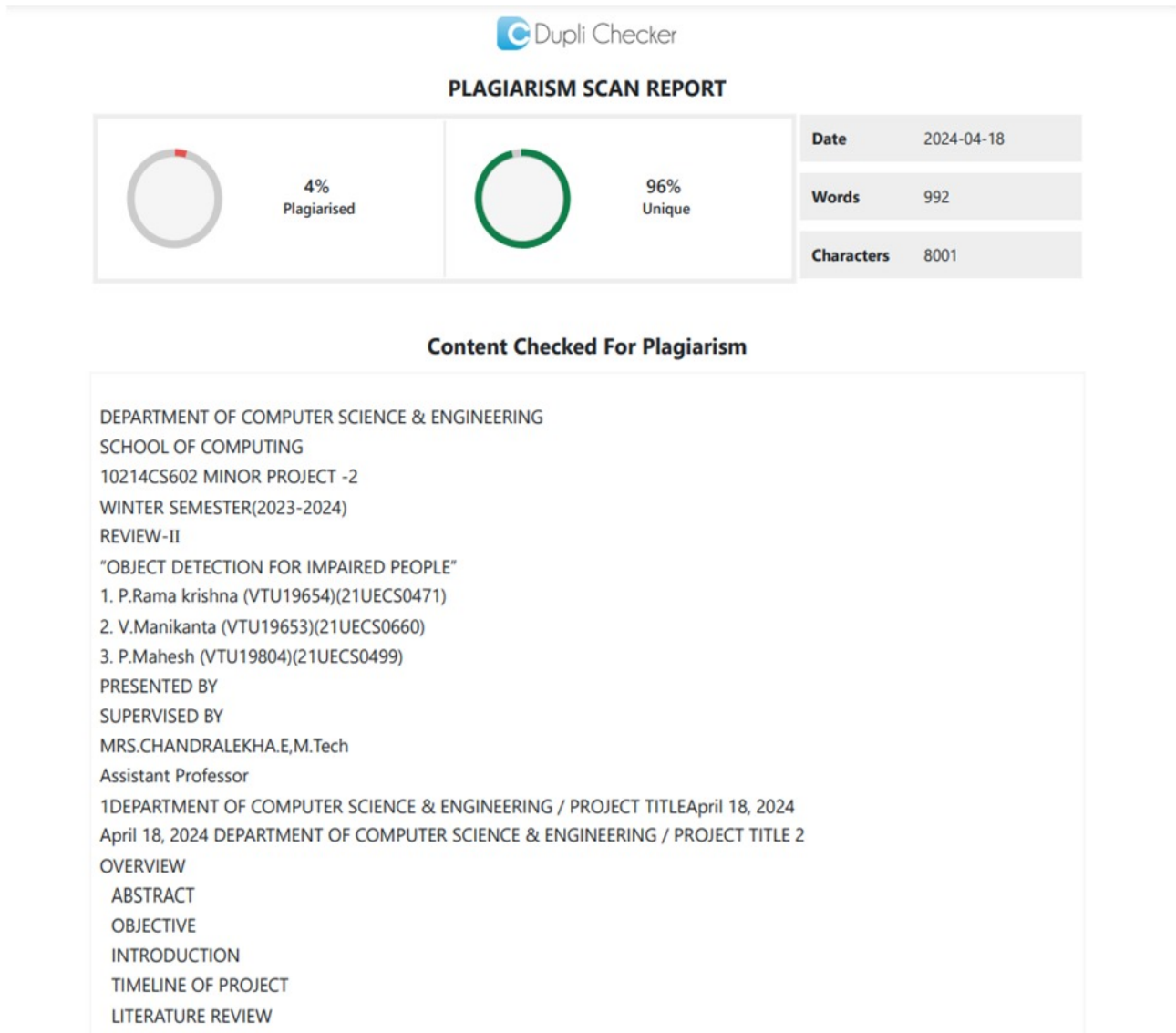


Figure 10.1: Plagiarism Report of Object Detection

Chapter 11

SOURCE CODE

11.1 Source Code

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import tensorflow as tf
5
6 from tensorflow import keras
7 from tensorflow.keras import layers
8 data_train_path = 'Fruits_Vegetables/train'
9 data_test_path = 'Fruits_Vegetables/test'
10 data_val_path = 'Fruits_Vegetables/validation'
11 img_width = 180
12 img_height = 180
13 data_train = tf.keras.utils.image_dataset_from_directory(
14     data_train_path ,
15     shuffle=True ,
16     image_size=(img_width , img_height) ,
17     batch_size=32 ,
18     validation_split=False)
19 data_cat = data_train.class_names
20 data_cat
21 data_val = tf.keras.utils.image_dataset_from_directory(data_val_path ,
22                                                         image_size=(img_height , img_width) ,
23                                                         batch_size=32 ,
24                                                         shuffle=False ,
25                                                         validation_split=False)
26 data_test = tf.keras.utils.image_dataset_from_directory(
27 data_test_path ,
28     image_size=(img_height , img_width) ,
29     shuffle=False ,
30     batch_size=32 ,
31     validation_split=False
32 )
33 plt.figure(figsize=(10,10))
34 for image, labels in data_train.take(1):
35     for i in range(9):
36         plt.subplot(3,3,i+1)
37         plt.imshow(image[i].numpy().astype('uint8'))
```

```

38         plt.title(data_cat[labels[i]])
39         plt.axis('off')
40 from tensorflow.keras.models import Sequential
41 data_train
42 model = Sequential([
43     layers.Rescaling(1./255),
44     layers.Conv2D(16, 3, padding='same', activation='relu'),
45     layers.MaxPooling2D(),
46     layers.Conv2D(32,3, padding='same', activation='relu'),
47     layers.MaxPooling2D(),
48     layers.Conv2D(64, 3, padding='same', activation='relu'),
49     layers.MaxPooling2D(),
50     layers.Flatten(),
51     layers.Dropout(0.2),
52     layers.Dense(128),
53     layers.Dense(len(data_cat))
54 ])
55
56 model.compile(optimizer='adam', loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=
    True), metrics=['accuracy'])
57 epochs_size = 25
58 history = model.fit(data_train, validation_data=data_val, epochs=epochs_size)
59 epochs_range = range(epochs_size)
60 plt.figure(figsize=(8,8))
61 plt.subplot(1,2,1)
62 plt.plot(epochs_range, history.history['accuracy'], label = 'Training Accuracy')
63 plt.plot(epochs_range, history.history['val_accuracy'], label = 'Validation Accuracy')
64 plt.title('Accuracy')
65
66 plt.subplot(1,2,2)
67 plt.plot(epochs_range, history.history['loss'], label = 'Training Loss')
68 plt.plot(epochs_range, history.history['val_loss'], label = 'Validation Loss')
69 plt.title('Loss')
70 image = 'Fruits-Vegetables/test/garlic/Image-2.jpg'
71 image = tf.keras.utils.load_img(image, target_size=(img_height, img_width))
72 img_arr = tf.keras.utils.array_to_img(image)
73 img_bat=tf.expand_dims(img_arr,0)
74 predict = model.predict(img_bat)
75 score = tf.nn.softmax(predict)
76 print('Veg/Fruit in image is {} with accuracy of {:.2f}'.format(data_cat[np.argmax(score)],np.
    max(score)*100))
77 model.save('Image_classify.keras')

```

```

1 import tensorflow as tf
2 from tensorflow.keras.models import load_model
3 import streamlit as st
4 import numpy as np
5 import pyttsx3
6
7 # Initialize the text-to-speech engine
8 engine = pyttsx3.init()
9 voices = engine.getProperty('voices')
10 engine.setProperty('voice', voices[1].id) # Select a female voice, adjust index as needed
11
12 st.header('Image Classification Model')
13
14 try:
15     # Attempt to load model
16     model_path = '\\Users\\pudur\\Downloads\\Image_classification\\Image_classify.keras'
17     model = load_model(model_path)
18 except Exception as e:
19     st.error(f"Error loading model: {e}")
20     st.stop()
21
22 data_cat = ['apple', 'banana', 'beetroot', 'bell pepper', 'cabbage', 'capsicum', 'carrot', 'cauliflower',
23             'chilli pepper', 'corn', 'cucumber', 'eggplant', 'garlic', 'ginger', 'grapes', 'jalepeno', 'kiwi', 'lemon',
24             'lettuce', 'mango', 'onion', 'orange', 'paprika', 'pear', 'peas', 'pineapple', 'pomegranate', 'potato',
25             'raddish', 'soy beans', 'spinach', 'sweetcorn', 'sweetpotato', 'tomato', 'turnip', 'watermelon']
26
27 img_height = 180
28 img_width = 180
29
30 image = 'corn1.jpg'
31 uploaded_file = st.file_uploader("Upload an image", type=["jpg", "jpeg", "png"])
32 if uploaded_file is not None:
33     image = uploaded_file.name
34
35 image_load = tf.keras.utils.load_img(image, target_size=(img_height, img_width))
36 img_arr = tf.keras.utils.img_to_array(image_load)
37 img_bat = np.expand_dims(img_arr, axis=0)
38 import numpy as np
39 import pandas as pd
40 import matplotlib.pyplot as plt
41 import tensorflow as tf
42 from keras import regularizers, layers, optimizers, utils, models
43 import pyttsx3
44 import streamlit as st
45

```

```

46 # Initialize the text-to-speech engine
47 engine = pyttsx3.init()
48 voices = engine.getProperty('voices')
49 engine.setProperty('voice', voices[1].id) # Select a female voice, adjust index as needed
50
51 st.header('Image Classification Model')
52
53 try:
54     # Attempt to load model
55     model_path = '\\Leela\\Image-classification (2)\\Image-classification\\Image-classify.keras',
56     model = models.load_model(model_path)
57 except Exception as e:
58     st.error(f"Error loading model: {e}")
59     st.stop()
60
61 data_cat = ['apple', 'banana', 'beetroot', 'bell pepper', 'cabbage', 'capsicum', 'carrot', 'cauliflower',
62             'chilli pepper', 'corn', 'cucumber', 'eggplant', 'garlic', 'ginger', 'grapes', 'jalepeno', 'kiwi', 'lemon',
63             'lettuce', 'mango', 'onion', 'orange', 'paprika', 'pear', 'peas', 'pineapple', 'pomegranate', 'potato',
64             'raddish', 'soy beans', 'spinach', 'sweetcorn', 'sweetpotato', 'tomato', 'turnip', 'watermelon']
65
66 img_height = 180
67 img_width = 180
68 image = 'corn1.jpg'
69 uploaded_file = st.file_uploader("Upload an image", type=["jpg", "jpeg", "png"])
70 if uploaded_file is not None:
71     image = uploaded_file.name
72
73 image_load = tf.keras.utils.load_img(image, target_size=(img_height, img_width))
74 img_arr = tf.keras.utils.img_to_array(image_load)
75 img_bat = np.expand_dims(img_arr, axis=0)
76
77 predict = model.predict(img_bat)
78 score = tf.nn.softmax(predict)
79
80 st.image(image, width=200)
81
82 prediction_text = data_cat[np.argmax(score)]
83 accuracy_text = ('With ') + str(round((np.max(score) * 100), 3)) + ' % accuracy '
84
85 st.write(prediction_text)
86 st.write(accuracy_text)
87 # Checkbox to toggle voice output
88 stop_voice = st.checkbox("Toggle Voice Output")
89 # Speaking the prediction
90 if not stop_voice:

```

```
91     try:
92         engine.say(prediction_text)
93         engine.runAndWait()
94     except RuntimeError:
95         pass # Handle RuntimeError gracefully
```

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